Community Detection on Signed Graphs: A Bayesian MCMC Coupling Approach

Seminar of the Neo team (Inria Sophia Antipolis)

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Inria Sophia Antipolis, Lagrange Gris room (L101) – 2.30 p.m. Thursday 25 June 2026

Outline

01
Motivation & the Bayesian framework

Haplotype Assembly

1. Biological Setup: Two Homologous Chromosomes (Haplotypes)

Humans are diploid: they have two copies of each chromosome (homologous chromosomes).

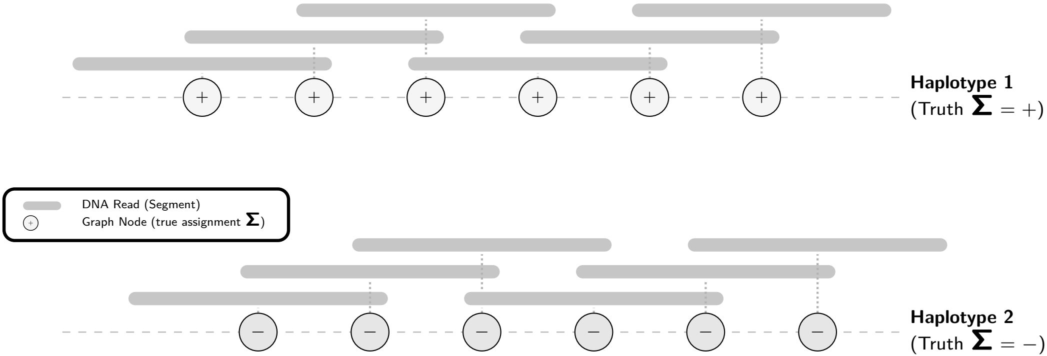
Haplotype 1
(Truth $\boldsymbol{\Sigma} = +$)

Haplotype 2
(Truth $\boldsymbol{\Sigma} = -$)

Haplotype Assembly

2. High-Throughput Sequencing: DNA Reads

Sequencing generates short, overlapping fragments called **reads** from both chromosomes.

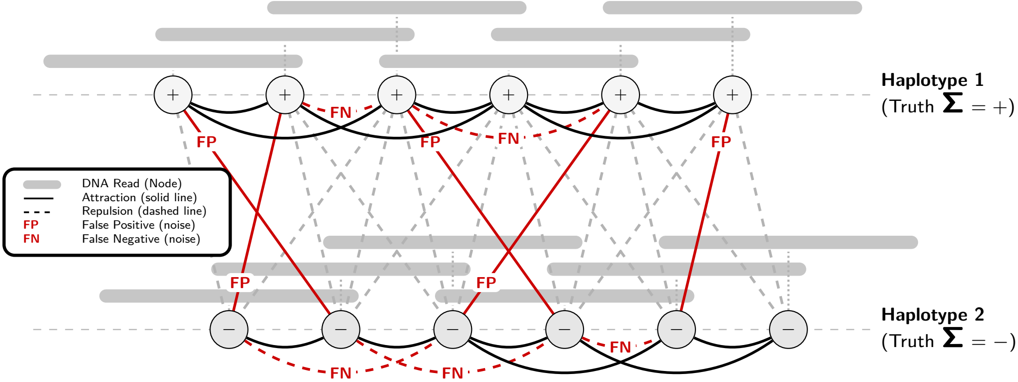


Haplotype Assembly

3. Modeling as a Signed Graph

Nodes = reads Edges = overlaps

Goal: Reconstruct the true haplotype assignments $\boldsymbol{\Sigma}$ from this observed signed graph.



Motivation: Signed Community Detection

Community Detection on Signed Graphs ($K = 2$)

- Partition nodes into $K = 2$ communities: $\Sigma_i \in \{+1, -1\}$.
- Links represent **attractions** (solid) and **repulsions** (dashed).
- Beyond the classical SBM: a rich framework coupling spatial geometry and community structure!

The Geometric Challenge

- Spatial embedding (*e.g.*, positions on $\mathbb{Z}$ or $\mathbb{R}^d$).
- Edge probabilities depend on distance $r = \|x_i - x_j\|$:

$$\mathbb{P}(\text{attractive edge} \mid \text{same community}) = f_{\text{in}}(r)$$
$$\mathbb{P}(\text{attractive edge} \mid \text{different communities}) = f_{\text{out}}(r)$$

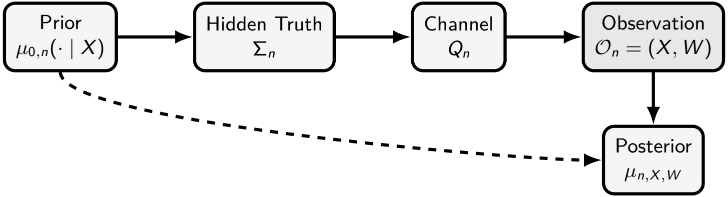
- **Main interest of the model... but also core challenge**: capturing this geometry.

The General Bayesian Model

Definition

1. **Nodes, Edges, Communities**: Nodes $V_n = \{1, \dots, n\}$, edges E_n, communities $\mathcal{L}_K = \{0, \dots, K-1\}$, and set of configurations $\sigma \in \mathcal{C}_n = (\mathcal{L}_K)^{V_n}$.
2. **Features**: $X_n \in \mathcal{X}_n$ according to distribution ν_n.
3. **Prior on configurations**: $\mu_{0,n}(\mathrm{d}\sigma \mid x)$ over $\mathcal{C}_n$.
4. **(Noisy) channel**: $Q_n(\mathrm{d}w \mid x, \sigma) = \prod_{e \in E_n} Q_{e,n}(\mathrm{d}w_e \mid x_i, x_j, \sigma_i, \sigma_j)$.

The joint law is $\mathbb{P}_n(\mathrm{d}x, \mathrm{d}\sigma, \mathrm{d}w) = \nu_n(\mathrm{d}x)\mu_{0,n}(\mathrm{d}\sigma \mid x)Q_n(\mathrm{d}w \mid x, \sigma)$.

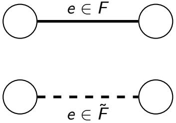


Gibbs Model on Signed Graphs

Attractive and Repulsive Edges

Partition potential edges $E = F \dot{\cup} \tilde{F}$:

- F: **Attractive** (ferromagnetic), weight $W_e > 0$
- $\tilde{F}$: **Repulsive** (antiferromagnetic), weight $W_e < 0$



Signed Gibbs Measure [1]

Configuration probability $\mu(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$, where energy $U(\sigma)$ penalizes unsatisfied edges:

$$U(\sigma) = \sum_{e=\{i,j\}\in F} W_e \mathbb{I}_{\{\sigma_i \neq \sigma_j\}} + \sum_{e=\{i,j\}\in \tilde{F}} |W_e| \mathbb{I}_{\{\sigma_i = \sigma_j\}}$$

Gibbs Assumption

Special case of the Bayesian framework where the posterior is assumed to be:

$$\mu_{X,W}(\sigma) = \frac{1}{Z} e^{-U(\sigma)}$$

Advantages of the Bayesian Framework

Rigorous Definitions

- **Algorithm** τ_n: Markov kernel mapping observations (X, W) to configurations.
- **Random Guess**: An algorithm agnostic to observations (relying only on the prior μ_0).
- **Weak / Partial Recovery**: Beating the best random guess asymptotically.

Posterior Resampling

Replace hidden truth by replica:
$\Sigma \iff \Sigma^{(1)} \sim \mu(\cdot \mid X, W)$

Measure of Performance: Overlap

The overlap quotients the Hamming similarity by label permutations $\mathfrak{S}_K$:

$$\mathrm{ov}_n(\sigma, \sigma') = \max_{\pi \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}_{\{\sigma_i' = \pi(\sigma_i)\}}$$

- Uniform prior: best random guess score is $1/K$.
- Weak recovery: overlap $\geq 1/K + \eta$.

Posterior Resampling (Nishimori Identity)

Core Idea: Replacing Truth by a Replica

Comparing τ to truth Σ $\iff$ Comparing τ to replica $\Sigma^{(1)} \sim \mu$

Theorem (Nishimori Identity [2])

For any algorithm τ, posterior replica $\Sigma^{(1)} \sim \mu$, and bounded test function Ψ:

$$\mathbb{E}\left[\Psi(X, W, \Sigma, \tau)\right] = \mathbb{E}\left[\Psi(X, W, \Sigma^{(1)}, \tau)\right]$$

Corollary (Overlap Identity)

$$\mathbb{P}\left[\mathrm{ov}(\Sigma, \tau) \geq s\right] = \mathbb{P}\left[\mathrm{ov}(\Sigma^{(1)}, \tau) \geq s\right] \quad (\forall s \in [0, 1])$$

02
The Sankararaman & Baccelli model

The Sankararaman–Baccelli Model (GSBM)

Model Definition [3]

- **Nodes**: Realization X of a homogeneous Poisson point process in an n-volume of $\mathbb{R}^d$ [4].
- **Latent Communities**: Uniform assignments $\Sigma_i \in \{+1, -1\}$ ($K = 2$).
- **Edges**: Connection $e = \{i, j\} \in H$ occurs with probability:

$$\mathbb{P}(e \in H \mid X, \Sigma) = \begin{cases} f_{\mathrm{in}}(r_e) & \text{if } \Sigma_i = \Sigma_j \text{ (attraction)}, \\ f_{\mathrm{out}}(r_e) & \text{if } \Sigma_i \neq \Sigma_j \text{ (repulsion)}, \end{cases}$$

where $r_e = \|X_i - X_j\|$ and $f_{\mathrm{in}}(r) \geq f_{\mathrm{out}}(r)$.

Goal: Weak Recovery

- Find an algorithm τ_n (independent of Σ_n) achieving:

$$\lim_{n\to\infty} \mathbb{E}[\mathrm{ov}_n(\Sigma_n, \tau_n)] \geq \frac{1}{2} + \eta \quad (\eta > 0), \quad \text{where } \mathrm{ov}_n(\Sigma_n, \tau_n) = \max_{\pi \in \mathfrak{S}_2} \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}_{\{\tau_{n,i} = \pi(\Sigma_{n,i})\}}.$$

GSBM in the Bayesian Framework

<table>
<tr><th>Sankararaman & Baccelli (GSBM)</th><th>Bayesian Framework</th></tr>
<tr><td>Positions X: Realization of a homogeneous Poisson point process</td><td>Features $X \sim \nu$: same Poisson law</td></tr>
<tr><td>Prior: Uniform prior μ_0 on $\Sigma \in \{+1,-1\}^n$</td><td>Prior μ_0: same</td></tr>
<tr><td>Observables: (H, X) where H is the observed graph</td><td>Observables: (W, X) where W is the channel output</td></tr>
<tr><td colspan="2">Deterministic construction of W from H:
$$W_e = \begin{cases} \log\left(\dfrac{f_{\mathrm{in}}(r_e)}{f_{\mathrm{out}}(r_e)}\right) \geq 0 & \text{if } e \in H \text{ (attractive),} \\ -\log\left(\dfrac{1-f_{\mathrm{out}}(r_e)}{1-f_{\mathrm{in}}(r_e)}\right) \leq 0 & \text{if } e \notin H \text{ (repulsive).} \end{cases}$$</td></tr>
<tr><td>Posterior: $\mathbb{P}(\Sigma = \sigma \mid H, X)$</td><td>Posterior: $= \mathbb{P}(\boldsymbol{\Sigma} = \boldsymbol{\sigma} \mid \boldsymbol{W}, \boldsymbol{X}) = \boldsymbol{\mu}(\boldsymbol{\sigma})$ (Gibbs measure)</td></tr>
</table>

From GSBM to the Gibbs Posterior

Conditioning on the Point Cloud

- Positions X are fixed $\implies$ all distances $r_e = \|X_i - X_j\|$ are known.
- Under a uniform prior, the posterior given the observed graph H satisfies:

$$\mathbb{P}(\Sigma = \sigma \mid H, X) \propto \mathbb{P}(H \mid X, \sigma)$$

Log-Likelihood as Signed Gibbs Energy

- Setting edge weights $W_e(H)$:

$$W_e(H) = \begin{cases} \log\left(\frac{f_{\rm in}(r_e)}{f_{\rm out}(r_e)}\right) \geq 0 & \text{if } e \in H \text{ (attractive)}, \\ -\log\left(\frac{1 - f_{\rm out}(r_e)}{1 - f_{\rm in}(r_e)}\right) \leq 0 & \text{if } e \notin H \text{ (repulsive)}. \end{cases}$$

- ... recovers the GIBBS posterior: $\mathbb{P}(H \mid X, \sigma) \propto e^{-U(\sigma)}$.

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The Percolation Obstruction

Necessary Condition for Weak Recovery [3]

- Global information flow requires the graph to **percolate** [5] (possess a giant component).

Theorem (Sankararaman & Baccelli [3])

Independent bond percolation with edge probability

$$p_e = f_{\mathrm{in}}(r_e) - f_{\mathrm{out}}(r_e)$$

yielding a giant component is a ***necessary condition*** *for weak recovery.*

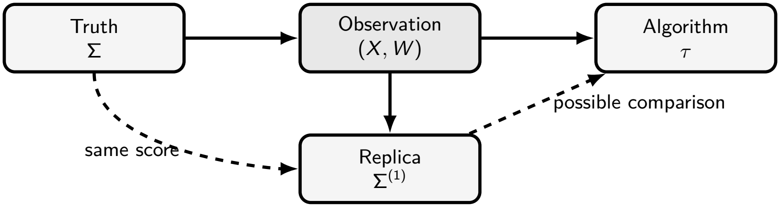
- **Our goal**: Retrieve and extend this result using unified **Bayesian MCMC coupling**.

03
MCMC Dynamics & Performance Bounds

Invariant Coupling

Markov Chain Dynamics as a Source of Coupling

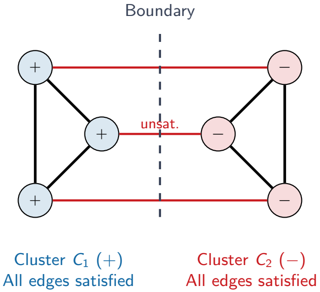
- We construct the posterior replica $\Sigma^{(1)}$ starting from the hidden truth Σ.
- Running **exactly one step** of a posterior-invariant $\textsc{Markov}$ kernel M initialized at Σ yields a valid replica $\Sigma^{(1)} \sim M(\Sigma, \cdot)$.



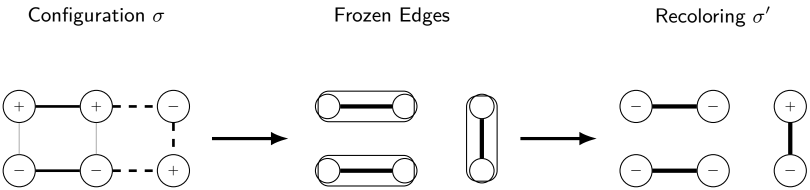
Single-Spin Dynamics & Geometric Bottleneck

Single-Spin Update Limitations

- **Geometry makes updates slow**: Single-spin dynamics (Glauber / Metropolis–Hastings [6]) update one node at a time.
- **Contradictory Clusters**: Two internally valid (highly satisfied) but mutually contradictory clusters are locked.
- **High Energy Barrier**: Flipping a single boundary node violates internal edges, creating a local energy barrier.
- **Conclusion**: Need for geometry $\implies$ **Cluster Dynamics** (updating whole regions).



One Step of Signed Swendsen–Wang [7, 8]



- Edges in bold represent satisfied interactions that are randomly frozen.
- Clusters are then flipped independently.

Swendsen–Wang Signed Cluster Dynamics

Swendsen–Wang (SW) Signed Cluster Dynamics [7, 8]

- Update whole clusters of vertices simultaneously to speed up mixing.
- **Freezing Step**: For each edge $e = \{i, j\}$:
  - If e is satisfied ($W_e > 0$ and $\sigma_i = \sigma_j$, or $W_e < 0$ and $\sigma_i \neq \sigma_j$), freeze it with probability:

$$p_e^{\text{frozen}} = 1 - e^{-|W_e|}.$$

  - If e is unsatisfied, do not freeze it ($p_e^{\text{frozen}} = 0$).
- **Recoloring Step**: Flip each frozen cluster independently with probability $1/2$.

High-Order Triangular Dynamics

The Frustration Obstruction

- **Frustration**: $\nexists \sigma$ satisfying all edges (conflicting constraints).
- **Excessive Freezing**: In frustrated configurations, standard SW satisfies and freezes too many edges independently, blocking/freezing the MCMC dynamics.
- **Higher-Order Correlation**: Group interactions into **triangles** to correlate freezing and **freeze as little as possible**.

Triangular Cluster Dynamics [9]

- **Energy Decomposition**: Distribute edge energy U over triangles: $U(\sigma) = \sum_{t \in \mathcal{T}} U_t(\sigma)$.
- **Freezing Mechanism**: Apply a joint freezing mechanism on the triangles (rather than independent edges) to keep cluster sizes active.

Triangular Dynamics Freezing Rules

△ **Attractive Triangle** (Isotropic)					
Configuration	$\varnothing$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+,+,+)$	b_u	0	0	0	a_u
Other	1	0	0	0	0
Repulsive Triangle (Isotropic)					
Configuration	$\varnothing$	$\{ab\}$	$\{ac\}$	$\{bc\}$	$\{ab, ac, bc\}$
$(+,+,+)$	1	0	0	0	0
$(+,+,-)$	b_u	0	$a_u/2$	$a_u/2$	0
$(+,-,+)$	b_u	$a_u/2$	0	$a_u/2$	0
$(+,-,-)$	b_u	$a_u/2$	$a_u/2$	0	0

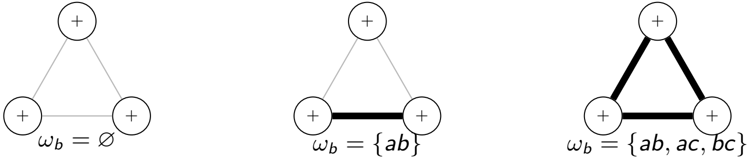
Table: Freezing rules for triangular dynamics ($a_u = 1 - e^{-2u}$, $b_u = e^{-2u}$), where u is the energy transferred by each edge.

General Swendsen–Wang Cluster Dynamics: Framework

General Framework [10]

Suppose the Hamiltonian decomposes as: $U(\sigma) = U_0(\sigma) + \sum_{b\in\mathcal{B}} U_b(\sigma)$, where each $b \in \mathcal{B}$ is a **bond** (*e.g.*, edge, triangle, clique).

1. **Freezing Step**: For each bond $b \in \mathcal{B}$, freeze a sub-bond $\omega_b \subseteq b$ with probability $P_b^\sigma(\omega_b)$.
2. **Recoloring Step**: Recolor each frozen cluster independently according to a distribution preserving the prior $\mu_0(\sigma) \propto e^{-U_0(\sigma)}$.



General Swendsen–Wang Cluster Dynamics: Invariance

Definition (Local Reversibility)

For two configurations σ, σ' and a bond b, we define the compatibility probability:

$$P_b^{\sigma \to \sigma'}(\omega_b) = \begin{cases} P_b^{\sigma}(\omega_b), & \text{if } P_b^{\sigma'}(\omega_b) > 0, \\ 0, & \text{otherwise.} \end{cases}$$

The freezing rule is **locally reversible** if for all b, ω_b, σ, σ':

$$P_b^{\sigma \to \sigma'}(\omega_b) = e^{-(U_b(\sigma') - U_b(\sigma))} P_b^{\sigma' \to \sigma}(\omega_b)$$

Theorem (Gibbs Invariance [10])

If freezing choices are independent, locally reversible, and $P_b^{\sigma}(\varnothing) > 0$ for all b, σ:

1. *The global cluster dynamics is reversible with respect to $\mu(\sigma) \propto e^{-U(\sigma)}$.*
2. *The condition $P_b^{\sigma}(\varnothing) > 0$ ensures the **irreducibility** of the MARKOV chain.*

Percolation as a Necessary Condition for Recovery

Theorem (Percolation Bound on Overlap [10])

Under the uniform prior μ_0, if frozen clusters are recolored independently and uniformly, then for the one-step posterior replica $\Sigma_n^{(1)}$, any algorithm τ_n, and any $\eta > 0$:

$$\lim_{n\to\infty} \mathbb{P}\left[\mathrm{ov}_n(\Sigma_n^{(1)}, \tau_n) \geq \frac{1}{K} + \frac{K-1}{K}(\theta^{\max} + \eta)\right] = 0$$

where $\theta^{\max}$ is the asymptotic fraction of nodes in giant frozen components (which bounds the maximum recoverable fraction of communities).

Corollary (Percolation is Necessary)

*If there is no percolation ($\theta^{\max} = 0$), the maximum overlap is bounded by $1/K$ (random guess): weak recovery is impossible. Hence, **percolation is a necessary condition for weak recovery**.*

Key Intuition

Small clusters are recolored independently, washing out their contribution. Only giant (macroscopic) components can carry global community information.

Coupling & the Sankararaman–Baccelli Bound

One Step of Classical Edge-by-Edge Swendsen–Wang

Let's run one step of standard SWENDSEN–WANG (edges) initialized at the hidden truth Σ. For any edge e, the probability that it is frozen is:

- **Case 1 ($\Sigma_i = \Sigma_j$)**:

$$\mathbb{P}(\text{frozen}) = f_{\text{in}}(r_e)\left(1 - e^{-|W_e|}\right) = f_{\text{in}}(r_e)\left(1 - \frac{f_{\text{out}}(r_e)}{f_{\text{in}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

- **Case 2 ($\Sigma_i \neq \Sigma_j$)**:

$$\mathbb{P}(\text{frozen}) = (1 - f_{\text{out}}(r_e))\left(1 - e^{-|W_e|}\right) = (1 - f_{\text{out}}(r_e))\left(1 - \frac{1 - f_{\text{in}}(r_e)}{1 - f_{\text{out}}(r_e)}\right) = f_{\text{in}}(r_e) - f_{\text{out}}(r_e)$$

Corollary (Retrieving the S&B Bound [3])

*Under the classical edge SW dynamics, the frozen graph is **exactly** an independent bond percolation with edge probability:*

$$p_e = f_{\text{in}}(r_e) - f_{\text{out}}(r_e).$$

Strict Improvement of the S&B bound (on the triangular grid)

Homogeneous Model ($f_{\text{in}} = p$, $f_{\text{out}} = 1 - p$)

On the triangular grid, we select a family of independent white triangles (like a chessboard).

- **Edge Dynamics (Standard SW)**: Frozen graph is independent edge percolation with parameter $p_{\text{edge}} = 2p - 1$. Critical threshold:

$$p_c^{\text{edge}} = \frac{1}{2} + \sin\left(\frac{\pi}{18}\right) \approx 0.673\ldots$$

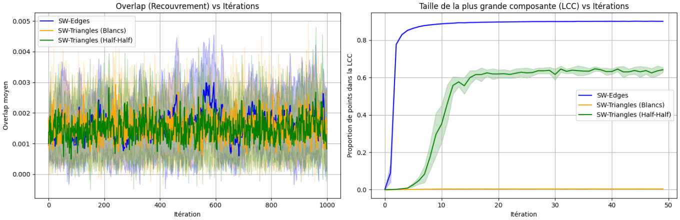
- **Triangular Dynamics**: Frozen graph yields correlated triangle percolation. Critical threshold:

$$p_c^{\triangle} = \frac{7}{4} - \frac{\sqrt{17}}{4} \approx 0.7192\ldots$$

Corollary (Strictly Better Impossibility Bound [10])

Since $p_c^{\text{edge}} < p_c^{\triangle}$, for any $p \in [0.673\ldots, 0.7192\ldots)$, the standard $\mathrm{S\&B}$ bound cannot conclude (edge dynamics percolate), whereas triangular dynamics do not, proving that weak recovery is impossible.

Empirical Results: Subcritical vs. Supercritical ($p = 0.70$)



Phase Space

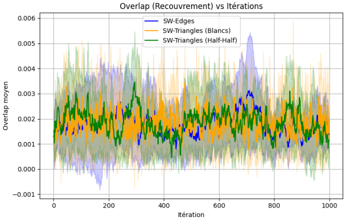
($p_c^{\text{edge}} \approx 0.673 < 0.70 < p_c^{\triangle} \approx 0.7192$)

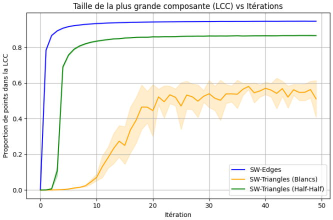
- **Edges (standard SW)**: supercritical ($\theta^{\max} > 0$), largest component size (LCC) reaches ≈ 0.9 (right plot).
- **Triangles / Split weight**: subcritical ($\theta^{\max} = 0$), LCC remains near 0.

Observed Behavior

- Only standard edge SW achieves giant components.
- Overlap remains at the baseline noise level (around 0.002) for triangles and split weights due to lack of percolation (left plot).

Empirical Results: Onset of Percolation ($p = 0.72$)





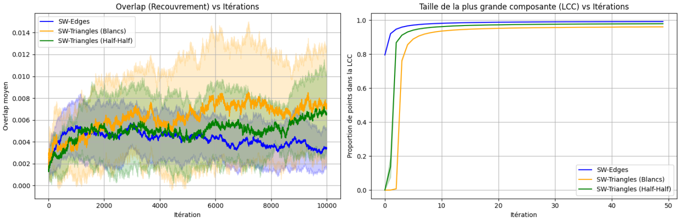
Transition Regime
($p = 0.72 > p_c^{\triangle} \approx 0.7192$)

- All three dynamics are now **supercritical**.
- LCC grows for all: edges reach ≈ 0.9, split weights reach ≈ 0.85, and white triangles reach ≈ 0.55 (right plot).

Observed Behavior

- Giant components have formed, but the overlap remains low (below 0.003) for all dynamics (left plot).
- **No weak recovery yet**: the true weak recovery threshold is likely strictly above the percolation threshold (percolation is only a necessary condition).

Empirical Results: Weak Recovery Achieved ($p = 0.80$)



Supercritical Regime ($p = 0.80 \gg p_c^{\triangle}$)

- Giant components are well-established (right plot), with LCC close to 0.95 for edges and split weights, and 0.90 for white triangles.
- Fluctuations are small and components are stable.

Observed Behavior

- Weak recovery is successfully achieved for all algorithms.
- The overlap climbs significantly above the noise level (left plot), peaking at ≈ 0.010 for white triangles and ≈ 0.006 for split weights.

04
Open Questions & Conclusion

Open Questions

Limits of the One-Replica Coupling

- Our framework constructs **one** replica $\Sigma_n^{(1)}$ using MCMC cluster dynamics.
- This only yields an impossibility bound (**necessary condition** for recovery).

Towards a Necessary and Sufficient Condition

- Achieving a **necessary and sufficient** condition requires a **two-replica** framework.
- Open: How to define and analyze two-replica couplings under MCMC?

Towards a Practical Community Detection Algorithm

Limits of Direct Gibbs Sampling

- Direct posterior sampling does not perform well in practice.
- Likely due to slow mixing of the underlying MARKOV chain.

Estimating the Correlation Matrix

- Estimating the correlation matrix $\mathbb{P}(\sigma_i = \sigma_j)$ yields much better results.
- **Future work**: Understand why correlation estimators are more robust.

Distance Adapted to Cluster Dynamics

Hamming Similarity on the Quotient Space

- Standard overlap is a HAMMING similarity quotiented by label permutations: $\mathrm{ov}_n(\sigma, \sigma') = \max_{\pi \in \mathfrak{S}_K} \frac{1}{n} \sum_{i=1}^{n} \mathbb{I}_{\{\sigma'_i = \pi(\sigma_i)\}}$.

Markov Transition-Adapted Distance

- Consider distances reflecting transitions of the cluster dynamics.
- For standard SW, this corresponds to the **symmetric difference** on satisfied edges:

$$d_{\mathrm{sym}}(\sigma, \sigma') = \frac{1}{2} \sum_{e=\{i,j\}} |W_e| \left| \mathbb{I}_{\{\sigma \text{ satisfies } e\}} - \mathbb{I}_{\{\sigma' \text{ satisfies } e\}} \right|$$

- Establishes links to **spectral graph analysis** (*e.g.*, LAPLACIAN eigenvalues).

Application: 3-SAT Solving via Signed Graphs

The 3-SAT Problem

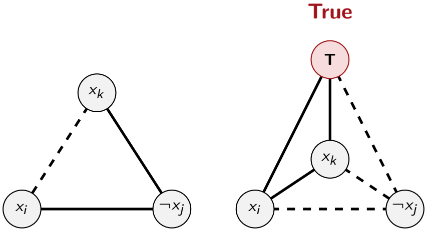
Find a truth assignment satisfying a set of clauses, *e.g.*,

$$\mathcal{C} = x_i \vee \neg x_j \vee x_k$$

Clause Encoding

We introduce a reference node **T** (True):

- **Triangle** (literals $x_i, \neg x_j, x_k$): edge signs opposite to the literal product.
- **Tetrahedron** (with **T**): edge signs conforming to the literal signs.



Triangular Bond (Base)

Tetrahedral Bond (with **T**)

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